

RYOTA MAEDA

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Research Interests

- Computer Vision
- Polarimetric Imaging
- Computer Graphics
- Light Transport Analysis
- Computational Imaging
- 3D Reconstruction

Work Experience

Pohang University of Science and Technology (POSTECH)

Apr. 2026 – Present

Postdoctoral Researcher

Education

University of Hyogo

Apr. 2022 – Mar. 2026

Ph.D. of Engineering

Supervisor: Prof. Shinsaku Hiura

University of Hyogo

Apr. 2020 – Mar. 2022

Master of Engineering

Supervisor: Prof. Shinsaku Hiura

University of Hyogo

Apr. 2016 – Mar. 2020

Bachelor of Engineering

Publications

Real-world Polarimetric Environment Map Dataset

Dec. 2026

Yonghee Oh*, **Ryota Maeda***, Yunseong Moon, Seung-Hwan Baek (**Equal contribution*)

SIGGRAPH Asia 2026

Transient Polarimetry

Jul. 2026

Oscar Pueyo-Ciudad, Guillermo Enguita-Lahoz, Yunseong Moon, **Ryota Maeda**, Seung-Hwan Baek, Albert Redo-Sanchez, Diego Gutierrez

SIGGRAPH 2026 Poster

Broadband Hyperspectral 3D Imaging using Dispersed Structured Light

Jul. 2026

Suhyun Shin, Yunseong Moon, **Ryota Maeda**, David B. Lindell, Kiriakos N. Kutulakos, Seung-Hwan Baek

SIGGRAPH 2026

End-to-End Optimization of Polarimetric Measurement and Material Classifier

Mar. 2026

Ryota Maeda*, Naoki Arikawa*, Yutaka No, Shinsaku Hiura (**Equal contribution*)

21st International Conference on Computer Vision Theory and Applications (VISAPP 2026)

Hyperspectral Polarimetric BRDFs of Real-world Materials

Dec. 2025

Yunseong Moon, **Ryota Maeda**, Suhyun Shin, Inseung Hwang, Youngchan Kim, Min H. Kim, Seung-Hwan Baek

SIGGRAPH Asia 2025

Event Ellipsometer: Event-based Mueller-Matrix Video Imaging

Jun. 2025

Ryota Maeda, Yunseong Moon, Seung-Hwan Baek

IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR), *Highlight*

Dense Dispersed Structured Light for Hyperspectral 3D Imaging of Dynamic Scenes

Jun. 2025

Suhyun Shin, Seungwoo Yoon, **Ryota Maeda**, Seung-Hwan Baek

IEEE / CVF Computer Vision and Pattern Recognition Conference (CVPR)

Polarimetric Light Transport Analysis for Specular Inter-reflection

May. 2024

Ryota Maeda, Shinsaku Hiura

IEEE Transactions on Computational Imaging, 2024

Refinement of Hair Geometry by Strand Integration

Oct. 2023

Ryota Maeda, Kenshi Takayama, Takafumi Taketomi

Computer Graphics Forum (Proceedings of Pacific Graphics 2023)

EpiScope: Optical Separation of Reflected Components by Rotation of Polygonal Mirror

Dec. 2021

Ryota Maeda, Shinsaku Hiura

SIGGRAPH Asia 2021 Technical Communications

Visiting Research Experience

POSTECH Computational Imaging Group Visiting Research <i>Mentor: Prof. Seung-Hwan Baek</i>	Mar. 2024 – Feb. 2025 <i>11 months</i>
CyberAgent AI Lab Research Intern <i>Mentors: Dr. Kenshi Takayama and Dr. Takafumi Taketomi</i>	Aug. 2022 – Sep. 2022 <i>2 months</i>
NAIST Optical Media Interface Lab Research Intern <i>Mentors: Prof. Hiroyuki Kubo and Prof. Yasuhiro Mukaigawa</i>	Aug. 2018 <i>1 month</i>

Software on GitHub

- Polanalyser** | ☆219 stars
Polarization image analysis tool. Demosaicing, Stokes vector, Mueller matrix.
- structuredlight** | ☆171 stars
Generate and Decode structured light. Binary, Gray, XOR, Ramp, Phase-Shifting, Stripe.
- EasyPySpin** | ☆113 stars
cv2.VideoCapture like wrapper for FLIR Spinnaker SDK.

Skills

- Programming:** Python, C++
- Hardware Development:** Electronic circuit design, Embedded systems, 3D CAD
- Visual Content Creation:** Photography, Image processing, Scientific illustration
- Language:** Japanese (native), English (advanced)